Compact arithmetic hyperbolic Coxeter polytopes: a dimension bound of 20

Coxeter Polytopes Frontier Project

20 October 2026; universal theorem manuscript

Abstract

We prove that compact arithmetic hyperbolic Coxeter polytopes have dimension at most20. A companion dimension and field reduction leaves only the actual ground field $\mathbb{Q}(\sqrt{5})$ in dimension21. The new ingredient is a normalized angle pool between distinct whole components of the same spherical type. Its total vertex cost is one nonnegative variable, while its local payments use actual component multiplicities. Complete contact-star coverage and D5/H4 incidence transfers permit an exact rational angle rule. Every allowed finite rank 21 vertex forest has charge at most 40. Independently evaluated native quadrilaterals have payment at least $60007431727/60000000000 > 1$; the inherited triangle payment is at least 2. A strict h-vector inequality forces average vertex charge greater than40, a contradiction. With the earlier bound21 and the thirteen-field reduction, this gives the dimension bound. The proof separates geometric coverage of actual faces from the finite rational checks.

1 Statement and scope

A Coxeter polytope is a finite-sided convex hyperbolic polytope with facet dihedral angles integral submultiples of π. Arithmeticity refers to its reflection lattice and its actual Vinberg ground field. It is not a condition on a convenient overfield containing displayed Gram entries.

Theorem 1.1. *Every compact arithmetic hyperbolic Coxeter polytope has dimension at most 20.*

The proof uses a preserved higher-dimensional reduction, the nongolden field arguments below, and a golden-field step. We state the inherited interfaces precisely so that their internal geometric proofs are not confused with the published field restrictions.

Proposition 1.2 (Companion dimension and field reductions)**.** *Every compact arithmetic hyperbolic Coxeter polytope has dimension at most21. If its dimension is21, its actual cyclic ground field is* $\mathbb{Q}(\sqrt{5})$.

The first assertion is the dimension-specific composition through the published compact ceiling29 in the preserved technical proof supplement `manuscripts/compact_arithmetic21.tex`. The second uses the dimension-at-least14 thirteen-field list and the twelve nongolden dimension21 exclusions stated in Section 7. Their proof homes and exact interfaces are given there; neither assertion is being imported as a published dimension20 theorem. The rest of this paper excludes the remaining golden-field case.

Published inputs, internal geometric arguments, and finite exact checks are distinct. Published assumptions include compact acute simplicity, finite/affine and Lannér classification, the arithmetic cyclic-product criterion and field restrictions, simplest-type reflection arithmeticity, rational

isotropy, and the nonnegative symmetric h-vector theorem [1, 2, 3]. The supporting-star reduction, whole-component normalization, incidence transfers and rational recurrence are internal proofs. Exact scripts certify explicitly finite obligations; a numerical optimizer only proposes a vector. The geometric completeness arguments precede the finite evaluations.

2 The strict counting threshold at rank 21

A planar angle is an incidence (v, F) of a vertex and a two-face. In a simple n-polytope it corresponds to an unordered pair of incident facet normals, marked in the finite vertex diagram.

Lemma 2.1. *Suppose nonnegative planar-angle weights on a compact simple21-polytope satisfy* $\sum_{v\in F} \mathrm{wt}(v, F) \geq 5-k$ *on every* k*-gonal two-face. Then the total vertex charge is at least* $40f_0+550$*, so its average is strictly greater than40.*

Proof. Each vertex is on $\binom{21}{2} = 210$ two-faces. Double counting gives

$$\sum_v \mathrm{wt}(v) \geq 5f_2 - 210f_0.$$

For the symmetric nonnegative h-vector of the simplicial dual,

$$f_0 = \sum_{i=0}^{21} h_i, \qquad f_2 = \sum_{i=0}^{21} \binom{21-i}{2} h_i.$$

Pair i with $21 - i$ for $0 \leq i \leq 10$. In $5f_2 - 210f_0$, their coefficient on $h_i = h_{21-i}$ is

$$5\left[\binom{i}{2} + \binom{21-i}{2}\right] - 420 = 80 + 5(i-10)(i-11) \geq 80.$$

The constant80 is their coefficient in $40f_0$. Every extra term is nonnegative, and $i = 0$ contributes $5 \cdot 10 \cdot 11\, h_0 = 550$ because $h_0 = 1$. Hence $5f_2 - 210f_0 \geq 40f_0 + 550$. □

It suffices to pay2 on triangles and 1 on quadrilaterals; nonnegativity handles $k \geq 5$. An upper certificate equal to 40 is enough for a contradiction. There is no need to manufacture a strict numerical margin.

3 Complete supporting-star coverage

We explain the geometric quantifiers needed for the new local weights. Use the Lorentz form of signature $(21, 1)$ and spacelike outward facet normals r_i with $(r_i, r_j) \leq 0$ for distinct walls. Chamber inequalities are $(x, r_i) \leq 0$.

Lemma 3.1. *Every positive-definite principal root set defines an actual face. A positive affine principal set is impossible in a compact chamber.*

Proof. Let B be the positive-definite principal Gram matrix and h a strict interior timelike vector. A positive-definite Z-matrix has entrywise nonnegative inverse, so $c = B^{-1}((h, r_i)) \leq 0$. The vector $h_0 = h - \sum_i c_i r_i$ is orthogonal to the chosen roots, has

$$h_0^2 = h^2 - ((h, r_i))^T B^{-1}((h, r_i)) < 0,$$

and has strictly negative product with every other facet normal. This gives a relative-interior point of the face. For an affine principal set, its positive kernel gives a nonzero null vector with nonpositive products against every wall. It cannot be zero because pairing its positive combination with h is negative. It gives an ideal point, contradicting compactness. □

Let W be the 19 supporting roots of a quadrilateral and b_0, b_1, b_2, b_3 its cyclic side roots. Every adjacent extension of W is elliptic. Each opposite extension is indefinite by Lemma 3.1: otherwise the corresponding opposite sides would meet on this face. Minimal indefinite subdiagrams contain both opposite black roots and are Lannér pieces. Complete initial union supports retain shared white roots, black identifications and every cross contact.

Lemma 3.2. *A k-root contact core containing a positive corner of order $k-2$ and a negative direction has nullity at most one. A complete quadrilateral star has nullity exactly one.*

Proof. The positive index is at least $k-2$ and the negative index is one. For the full star $k = 23$ but the ambient Lorentz dimension is22, so it is singular and the nullity is exactly one. □

3.1 The finite seed and extension domains

A contact state records the four cyclic side roots, the supporting white roots already entered, all native labels, each fixed dotted parameter, and the exact Gram data. Isomorphisms must preserve the four marked roles and are accompanied by a checked diagonal gauge congruence. A state is unpaid when no proved persistent part of the baseline rule has already supplied a unit.

Proposition 3.3 (Finite parameter domains)**.** *Every unpaid actual golden-field quadrilateral starts in one of the opposite-pair union domains. At each state its next touching white root belongs to a finite exact one-white domain. The dotted parameters in the seed domains have finite integral bounds that remain valid in every larger principal extension.*

Proof. The finite-angle alphabet has no cutoff hypothesis. The two-cycle $4\cos^2(\pi/m) = 2 + 2\cos(2\pi/m)$ belongs to $\mathbb{Q}(\sqrt{5})$; its real cyclotomic degree is at most two. The candidates are $2, 3, 4, 5, 6, 8, 10, 12$, and $8, 12$ generate the wrong quadratic fields. Thus exactly $2, 3, 4, 5, 6, 10$ remain. For a three-node piece, enumerate the 56 unordered triples. Its nonzero cycle belongs to the ground field precisely when the counts of labels $4, 6, 10$ are even: the square classes of $2, 3, \varphi + 2$ are independent. Here $2, 3, 6$ are not squares in $\mathbb{Q}(\sqrt{5})$, and the norm of $\varphi + 2$ is five. A right triangle has zero cycle and needs no parity test. Exact physical determinant negativity and conjugate determinant positivity leave the twelve listed triangle types in the attached catalogue. At order four or five, a bond at least six with a neighboring nonright bond gives a proper three-node determinant at most $1 - 3/4 - 1/4 = 0$; a dotted pair is already a nonpositive proper pair. Thus the bonds are $3, 4, 5$. Any cycle must span all nodes and have no chord: a shorter all-three cycle has a nonpositive all-ones witness. Consequently enumerate all labeled trees (for example by Prüfer words), all spanning chordless cycles, and all assignments $3, 4, 5$, then test every proper principal matrix, physical negative index, field cycles and conjugate positivity. Quotienting only by explicit label-preserving maps leaves six order-four and four order-five records. An order-two dotted piece is retained symbolically as $t \in \mathbb{Z}[\varphi]$, $t > 4$, $0 < t' < 4$ until the bound below. This defines the complete catalogue test used here; it does not assert that a passing local diagram extends to a chamber. The remaining radicands $2, 3, \varphi + 2$ are totally positive. Once cycle products are fixed, alternative real radical signs differ by diagonal root sign changes, so conjugate positivity is a well-defined test independent of an arbitrary choice of displayed root signs.

For each opposite pair, a minimal indefinite subset contains both black roots, since deleting either leaves a subset of an actual finite corner. An affine principal subset is impossible by Lemma 3.1. The minimal pieces are therefore Lannér diagrams of order at most five. Enumerate their allowed white identifications and the remaining contacts in the union. Every unfixed pair apart from the two opposite pairs belongs to an actual finite corner. Its label belongs to the golden native alphabet $2, 3, 4, 5, 6, 10$; no dotted distance is sampled in filling those pairs. Exact finite-corner and arithmetic tests give a finite initial domain once the opposite dotted parameters have been bounded.

Here is the parameter bound used in the remaining24 minimal dotted families. Their exact inertia certificates identify a common block B, obtained by deleting the dotted endpoints0 and 2, that is nonsingular and already has one negative direction. If the dotted entry is $-x$ and the endpoint squares are d_0, d_2, its Schur residual has the form

$$\begin{pmatrix} a & -x-c \\ -x-c & b \end{pmatrix}.$$

The residual must be positive semidefinite: the common block has already used the permitted negative direction. Thus

$$(x+c)^2 \le ab, \qquad t := \frac{4x^2}{d_0 d_2} \le \frac{8(ab+c^2)}{d_0 d_2}.$$

This is a principal-matrix obstruction and survives every extension. The exact ceilings U of the displayed upper bounds lie between9 and 76 in the finite family table. Cyclic integrality gives $t = A + B\varphi \in \mathbb{Z}[\varphi]$, where $\varphi = (1+\sqrt{5})/2$. The physical and conjugate conditions give

$$4 < t \le U, \qquad 0 \le t' \le 4.$$

Hence $B = (t-t')/\sqrt{5}$ is positive and at most U. Also $A = (\varphi t' - \varphi' t)/\sqrt{5}$ lies between0 and U: use $U \ge 4$ and $\sqrt{5} - |\varphi'| = \varphi$ for the upper bound. The integer box $0 \le A \le U$, $0 < B \le U$ is therefore complete. Every point is tested for the exact field and gauge, cycle integralities, physical inertia and conjugate positivity. The recorded evaluation gives532 parameter candidates and 32 necessary surviving seeds. Passing these tests is not an assertion of geometric realization.

Finally, an added white root belongs to all four actual finite corners. Its products with entered roots are finite native labels from the same alphabet. Enumerate all label tuples with at least one nonright contact to the current core, retaining exactly those whose four enlarged corners are finite and whose exact arithmetic, root relations and embedding inertias pass the necessary tests. Equivalently, generate the tuples by complete finite-Dynkin attachment. The old dotted invariant stays fixed; a new supporting white introduces no free dotted parameter. This proves finiteness and the required coverage of a single extension. □

The finite Lannér family table, minimal-union enumeration and the 24 exact Schur bounds are internal certificate inputs. Their full domains and independent checks are in `results/fresh_sqrt5_quad_exhaustion_20261015.md` and `results/fresh_sqrt5_dotted_parameters_20261015.md`. The argument above explains why those finite tables suffice; checking their hashes alone would not establish that reduction. Here is the data interface used by those inputs. A diagram record has its ordered nodes, the lexicographically ordered array of pair labels, the four fixed black roles, and the white subset. Its exact Gram normalization records the diagonal root squares and field entries; any alternate labeling carries an explicit mark-preserving diagonal gauge congruence. A dotted-family record additionally gives the deleted endpoint indices, the exact matrix B, its inertia pivots, the Schur entries a, b, c, d_0, d_2, and the integral ceiling U proved by their displayed inequality. All

numbers are rational pairs in the integral field basis or specified totally real radicals, not decimal eigenvalues. The finite table is checked by exact proper-principal positivity, physical negative index, cycle integrality and conjugate positivity. The independent minimal-union replay uses finite-Dynkin recognition and separate literal white-identification generation. These finite classified tables are attached proof data, while the all-extension coverage argument is Proposition 3.5 below.

3.2 The contact queue and its stopping obligations

The algorithm begins with every retained marked seed and preserves its literal seed map. Process each new representative once. Record every singular state as a stopping obligation, including a singular state that also has accepted children. Otherwise generate the entire one-white domain of Proposition 3.3. A branch may be discharged by a uniform proof of persistent payment, rejected by a necessary-condition failure, or retained with its exact marked child map. A conditionally sealed unpaid state permits only orthogonal spectators. At31 roots retain a singular obligation and reject a nonsingular core by rank.

Proposition 3.4 (Termination)**.** *The contact queue is finite, independently of a time or memory limit.*

Proof. A k-root core has a positive corner of order $k-2$ and a negative direction, so its rank is at least $k-1$. The published compact ceiling29 allows ambient Lorentz rank at most 30. Thus at most 31 roots can occur; at31 only a singular core can remain. There are finitely many seed parameters, finitely many native label tuples at each extension, and a uniform bound on the number of extensions. Consequently even the labeled state tree is finite. Passing to representatives by verified marked isomorphisms preserves this property. Each representative is processed once, so the algorithm halts. Its completeness still requires the actual-face argument below. □

Proposition 3.5 (Final touching-star principle)**.** *For an unpaid actual quadrilateral, the inherited complete finite/dotted contact queues contain a final contact representative whose marked components are entire. Its remaining supporting components are common orthogonal spectators. Singular states that can extend must also remain stopping obligations.*

Proof. Begin with the actual opposite-pair minimal union. In each white component touching the entered core, order unentered vertices starting with a contact to that core and continuing along connected white edges. Every next root belongs to its complete one-white contact domain. Rejected arithmetic, signature or conjugate profiles cannot occur. A profile producing persistent basic payment pays the face independently of all adjustable bonuses. Otherwise a literal retained child map applies.

At a conditionally sealed state any further touching contact would give such persistent payment; hence the unpaid actual face has only orthogonal spectators. At an unsealed singular state continue while actual touching roots remain. Choose the final such state, rather than an earlier embedded singular subcore. Its contact part has corank one after the positive spectator block is split off, by Lemma 3.2. It has at most 23 roots. The complete queues retain every singular stopping state, including expandable and boundary states, within their larger geometric root bound. Its marked components are therefore whole in the actual corners. The remaining components are orthogonal to the black and entered white roots. □

Propositions 3.3–3.5 separate three obligations: finite extension domains, termination, and geometric coverage by final representatives. The recorded finite queue has5172 states and 51599 contact profiles; its2070 singular obligations include expandable states. The dotted queue has438 states,5146

profiles and 264 singular obligations. All423 dotted states below31 roots have complete extension domains. These are computed outputs of the stated algorithm. The exact maps and the independent queue evaluations are documented in `results/fresh_sqrt5_nonsingular_rank29_fast_20261015_review.md` and `results/fresh_sqrt5_dotted_rank29_exhaustion_20261015_review.md`. The final rational evaluator reads this saved domain; it does not regenerate either queue.

The rank 21 native iterator combines the prior complete parameter-stress inventory with every fitting singular finite-queue state. There are1979 native faces. For an N-root contact row, common spectator rank is $23 - N$. Enumerate exact external counts (e, f, g, j) of $A_1, I_2(5), H_3, A_4$ with

$$e + 2f + 3g + 4j \le 23 - N, \qquad 23 - N - e - 2f - 3g - 4j \ne 1.$$

There is no untracked rank-one type; A_2, A_3 realize every residual rank at least two. The same spectator forest is used at all four corners. This gives102917 cases. Exact component multiplicities determine all denominators. A numerical payment from an embedded old subcore is not substituted for the actual new angle.

4 The angle rule and its geometric exceptions

Write $q = (a, b, h, d)$ for the exact numbers of entire $A_1, I_2(5), H_3, A_4$ components. Old coefficients use the capped selector $c = (a, \min(b, 2), \min(h, 1), \min(d, 2))$. The fixed baseline on a whole connected component C is

$$B(C; i, j) = \mathbf{1}_{\mathrm{dist}_C(i,j)\le 2} + \mathbf{1}_{C=H_4}\mathbf{1}_{\{\mathrm{dist}_C(i,j)=3\} \text{ or } \{ij \text{ is the five-edge}\}}.$$

Lemma 4.1 (The baseline triangle payment)**.** *Every triangular two-face has baseline payment at least two, including arbitrary additional supporting roots.*

Proof. A minimal indefinite principal subset of its star contains all three black roots: omitting any one leaves an actual positive corner. Compactness excludes an affine principal subset. The complete golden Lannér table therefore gives $12 + 6\binom{4}{3} + 4\binom{5}{3} = 76$ three-black markings. The exact finite check takes each catalogue graph, each black triple, and each omitted black root, and computes the distance and entire H_4 bonuses in the remaining positive graph. Sixty-nine markings already have distance-two total at least two. Here is the entire deficient list, with numbered vertices $0, \dots, 4$; unlisted bonds are right. Corner triples are in increasing order of the omitted black.

nonright edges	black triple	baseline corner weights
$04 = 14 = 34 = 3,\ 23 = 5$	012	$(1, 1, 1)$
$04 = 13 = 14 = 3,\ 23 = 5$	023; 024	$(2, 0, 0); (1, 1, 0)$
$04 = 4,\ 13 = 14 = 3,\ 23 = 5$	023; 024	$(2, 0, 0); (1, 1, 0)$
$04 = 23 = 5,\ 13 = 14 = 3$	023; 024	$(2, 1, 0); (1, 2, 0)$.

The catalogue and these literal markings, not sampled wall distances, are the finite data checked in the triangle certificate. Every path already present remains the unique path in a finite spherical extension, so its distance cannot increase. A finite component containing an induced H_4 is that whole H_4, by the finite classification of diagrams with a 5 bond. Thus each extra unit used in the table survives every additional white root. Disconnected marked nodes can only gain a nonnegative payment. This proves the full-star assertion. No extra $I_2(5)$ or H_3 five-edge payment is needed for this lemma. □

We now define every coefficient orbit of the additional rule. A rooted type means a finite labeled connected diagram with one distinguished node, modulo label-preserving isomorphism; a pair type has an unordered distinguished node pair. The data give a nonnegative rational value $v_c(\mathfrak{o})$ for each orbit and capped context c; unlisted values are zero. For an actual marked angle use this exhaustive, mutually exclusive selector:

1. If both marks lie in one component, add its pair-type value only when its baseline is zero. There is no multiplicity denominator.

2. Otherwise order the first three hub families as $A_1 < I_2(5) < H_3$. If one marked component is a hub and the other is not in that family or any earlier hub family, use the first such hub H as source. The orbit records H, the destination family and its rooted type, and the addition is $v_c(\mathfrak{o})/(\mathrm{rk}(H)k_H)$, where k_H is the exact number of whole H components. It is independent of the selected source root. Thus the denominators are $a, 2b, 3h$, and an A_1–H_3 angle, for example, belongs only to the A_1 source family.

3. If no preceding selector applies and both components are A_4, use their unordered pair of rooted types and divide its value by $4(d-1)$; this branch exists only for $d \geq 2$.

4. If exactly one component is A_4 and the other is none of the four hub types, the orbit records both rooted types, with A_4 first, and the addition is its value divided by $4d$.

5. All remaining cross-component angles have no old bonus.

This defines a literal angle from its whole finite vertex diagram and the attached rational coefficient table, without calling an optimizer. Rooted and pair orbits can be encoded recursively by the labeled tree rooted at its center, recording at each node its mark and the sorted list of labeled child codes; two-center trees use the unordered two halves and their joining label. The attached variable-order table identifies these codes and contexts. Numeric column indices are merely serialization, not part of the mathematical rule.

At fixed c these old charges are additive over components. For each destination root, summing a hub source over all $\mathrm{rk}(H)k_H$ source roots cancels its denominator. An A_4 source similarly contributes one fourth of the sum over its four rooted types to each destination root. For distinct A_4 factors, summing over factor pairs gives $\binom{d}{2}[4(d-1)]^{-1}\sum_{s,t\in A_4} v_c(\{s,t\}) = (d/8)\sum_{s,t\in A_4} v_c(\{s,t\})$. These identities define the additive type costs $w_c(T)$ used below.

4.1 D5 and H4 incidence transfers

These internal geometric lemmas are used with their complete written proofs and exact coefficient certificates.

Proposition 4.2 (Previously proved priority rules)**.** *Let W denote the old total charge and b the old extreme-E_8 pair bonus. There is a nonnegative angle rule preserving all required small-face payments with the following conservative allowances:*

Whole vertex forest	*Allowance*
$D_5 + 2E_8$	$W - \min(1, 2b)$
$A_1 + H_4 + 2E_8$	$W - t$
$A_1 + A_4 + 2E_8$, $A_5 + 2E_8$	$W + t$
$A_1 + B_4 + 2E_8$, $B_5 + 2E_8$	$W + t$
All other forests	W

Here $0 \le t \le 1$. The two D5 state inequalities must both be imposed.

We prove both priority rules, including their simultaneous consumers. All roots in the local calculations below have square two unless a label-four scaling is specified. Put $s = \sqrt{5}$, $\varphi = (1+s)/2$ and $\mathcal{O} = \mathbb{Z}[\varphi]$. A whole untouched E_8 is orthogonal to any new side seen in a finite corner: it has no connected finite extension. Thus its roots really split off in each calculation.

4.1.1 The triangle activity and the D5 fork deletion

Number D_5 by edges $01, 12, 23, 24$, with short leaves $A = 3, B = 4$. The complete necessary nonadjacent triangle domain has the following third-wall contacts; all unspecified products are zero:

old type	contact magnitudes	marked triangles
D_5	$\varphi@0$	$(0,2)$
D_5	$\varphi@f$ or $1@1 + \varphi@f,\ f = 3,4$	$(1,f)$
E_8	$\varphi@0$	$(0,2)$
E_8	$\varphi@6$	$(3,5), (3,6), (4,6).$

Here E_8 is the path $0 - 1 - \cdots - 6$ with 7 attached at 2. For clarity, this is a finite exact domain calculation: for each nonadjacent marked pair generate every finite one-node extension of the corner with one mark deleted, then give the other mark each contact $2, 3, 4, 5$ and test the other corner. A bond at least six to any old node would have to be an isolated $I_2(m)$ in every corner, impossible because an old neighbor remains. Test positive physical height $H = g^T J^{-1} g - 2$, cycle membership in $\mathbb{Q}(s)$, conjugate $H' \le 0$, and all affine principal subsets. These literal finite tests leave exactly the displayed list; the complete eligible vectors and exact tests are supplied with the triangle-domain data. No high-label cutoff or geometric realizability filter is imposed. The D_5 fork pair is absent, so it is never triangular. The two E_8 root heights are $4 + 2s$ and $1 + s$. Let I_E indicate the occurrence of any of these four triangles in either whole E_8; there are eight designated possible triangles.

Consider a fork quadrilateral with cyclic sides A, B, c, d and white $W = \{0, 1, 2\}$. If it has a basic unit not being deleted, it is paid. Otherwise $c \cdot d = 0$, and finite extension of the endpoint A_4 permits precisely

$$N, \quad P_{03}, P_{04}, P_{13}, P_{23},$$

where N is no contact and P_{ij} is contact at white i with label j. A contact to the adjacent old black root would already be a retained adjacent-pair unit. A rank-five finite extension of A_4 is A_5, B_5, D_5, so there are no other endpoints or high bonds. Testing $W + c + d$ as a finite tree leaves exactly the eleven unordered pairs in the next table. All excluded pairs have a cycle, degree-four vertex, or a branched finite diagram with an internal four-bond.

The all-distance arithmetic reduction is short enough to give explicitly. A label-three side can be field-scaled to square two and contact -1; its old three-path cycle makes its opposite product an element of $\mathcal{O}$. A label-four side is instead scaled to square four and contact -2; the Cartan cycle still makes its opposite product integral, with no lost factor of two. In that cycle the two incident factors are -2 and $x/2$; the old label-three path contributes signs only, leaving $\pm x$. For an N side use the invariant opposite square $X \in \mathcal{O}$, without assuming its norm-two root field-valued. The complete finite opposite lists, subject to the strict opposite-corner inequalities, are

$$x = -2\cos(\pi/m) \in \mathbb{Q}(s): \quad -1, -\varphi; \qquad X > 5/2: \quad \varphi^2, 3, (5+s)/2.$$

A square-four side has only the finite field-valued opposite value -2. Indeed the two-cycle restricts the labels to $2, 3, 4, 5, 6, 10$; other values require $\sqrt{2}, \sqrt{6}$ or $\sqrt{5+s} \notin \mathbb{Q}(s)$. The same cyclotomic restriction follows from degree at most two of $4\cos^2(\pi/m)$, so it is not a sampled list.

Eliminating W uses $J_W^{-1} = \frac{1}{4}\begin{pmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{pmatrix}$. For the following notation let

$$\begin{aligned} f_0(X,y) &= (5-2X)(3+y-2y^2)+3y-7, \\ f_1(X,y) &= (5-2X)(1+y-y^2)+3y-5, \\ f_2(x,y) &= (2x-3)((x+1)y^2-(x+2)y)-2(x-1)^2, \\ f_3(x,y) &= (2y+1)(3+x-2x^2)+3x-7, \\ f_4(x,y) &= (2y+1)(1+x-x^2)+3x-5. \end{aligned}$$

Corank one gives exactly these equations and the three unit equations:

Endpoints	corank equation	disposition
N, N	$(2X-5)(2Y-5)=9$	Lambert branch below
N, P_{03}	$f_0(X,y)=0$	excluded
N, P_{04}	$f_1(X,y)=0$; square-four y	excluded
N, P_{13}	$f_1(X,y)=0$	excluded
P_{03}, P_{13}	$f_2(x,y)=0$	excluded
N, P_{23}	$(5-2X)(2y+1)=9$	unit branch below
P_{03}, P_{23}	$f_3(x,y)=0$	excluded
P_{04}, P_{23}	$f_4(x,y)=0$; square-four x	excluded
P_{23}, P_{23}	$(2x+1)(2y+1)=9$	excluded
P_{03}, P_{03}	$(x+1)(y+1)=1$	unit branch below
P_{13}, P_{23}	$f_4(y,x)=0$	excluded.

These are exact Schur determinants, not assumptions about opposite lengths. Each excluded row forces at least one opposite pair finite. For f_0, solving for X gives $X-4=(3y^2-8)/(3+y-2y^2)<0$ if $y\le -2$. For f_1 the denominator is $2(1+y-y^2)$ with the same numerator. For f_3, f_4, solving for y gives $y+2=(8-3x^2)/(3+x-2x^2)$ or $(8-3x^2)/(2(1+x-x^2))>0$ in the dotted range. For f_2 at $x, y\le -2$, its left side is at least $f_2(x,-2)=2(5x^2+x-13)\ge 10$. Thus there is no unbounded opposite-distance search. Substitution of the complete finite lists gives all the obstructions:

Row	nonintegral value or nonsquare discriminant
N, P_{03}	$y=-\varphi$ gives $X=(25+12\varphi)/11$; finite X gives $134\pm 50s, 84$, norms 5456 or rational nonsquares.
N, P_{13}	$y=-1,-\varphi$ gives $13/2, 2+5\varphi/4$; finite X gives $26-6s, 28, 34+14s$, norms $496, 784, 176$; 28 is neither a rational square nor five times one.
N, P_{04}	finite square-four $y=-2$ gives $X=18/5$; finite X has the preceding discriminants.
P_{03}, P_{13}	$x=-\varphi$ gives $73+89\varphi$ of norm 3905; $y=-1,-\varphi$ gives $104, 153+224\varphi$, norm 7505 for the latter.
P_{03}, P_{23}	$x=-\varphi$ gives $y=-(3+12\varphi)/11$; $y=-1,-\varphi$ gives discriminants $21, 134+50s$, the latter of norm 5456.
P_{04}, P_{23}	$x=-2$ gives $y=-8/5$; $y=-1,-\varphi$ gives $28, 34+14s$, the latter of norm 176.
P_{13}, P_{23}	finite $y=-1,-\varphi$ gives $x=-9/2, -5\varphi/4$; finite $x=-1,-\varphi$ gives $28, 34+14s$.

For a nonrational square in $\mathbb{Q}(s)$ its rational norm must be a rational square; a rational number must be a square or five times a square in $\mathbb{Q}$. These tests exclude every listed branch. In the P_{23}, P_{23} row, both dotted implies $x=y=-2$, whose rational indefinite Gram violates conjugate positivity. Finite $x=-1$ gives $y=-5$, with rational opposite principal determinant -21; $x=-\varphi$ gives $y=-1/2-9s/10\notin\mathcal{O}$. This also excludes its exchanged cases. In particular the mixed P_{13}, P_{23} row is impossible: no fixed D5 half-units are being silently added to the golden baseline.

Here are the complete unit branches. Every positive unit of $\mathcal{O}$ is φ^r: scale it into $[1,\varphi)$ and use its norm ±1 and the fact that its conjugate difference is an integer multiple of s. Norm -1 makes r odd. Both 2 and 3 are inert, since t^2-t-1 has no root modulo either prime. For N,N, set $U=2X-5, V=2Y-5$. Opposite indefiniteness and conjugate positivity give

$$UV=9,\quad U,V>0,\quad -5\le U',V'\le -9/5,\quad U,V\equiv 1 \pmod 2.$$

Thus $U=3^a\varphi^r$, $a=0,1,2$, $r\equiv 3 \pmod 6$. The conjugate interval leaves only $(a,r)=(0,-3),(2,3)$: neighbors differ by a factor $\varphi^6>16$ and are already outside. Hence $\{X,Y\}=\{\varphi^2,(23+9s)/2\}$. For N,P_{23} use $U=2X-5, V=-2y-1$ with the same conditions. The two branches are $(X,y)=(\varphi^2,-(19+9s)/2)$ and $((23+9s)/2,(1-s)/2)$; the latter has a forbidden opposite product in $(-1,0)$. In the former a side has just the short-leaf contact $-\varphi$ at each of its two fork corners. For P_{03},P_{03} put $U=-x-1, V=-y-1$. Now $UV=1$, $U,V>0$, and conjugate positivity gives $-5/2\le U',V'\le -2/5$. Only $r=\pm1$ remains, so $\{x,y\}=\{-\varphi,-\varphi^2\}$.

These three surviving necessary patterns have no E_8 triangle at any fork corner. For the first two, the short-leaf side height is $H_L=(-1+5s)/8$; its length-two cyclic products with the two E_8 triangle walls would be $(23+9s)/4$ and $3+s/2$, not in $\mathcal{O}$. For the third, its side has height $H=(1+9\varphi)/4$ and the corresponding products are $(19+29\varphi)/2,(9+10\varphi)/2$, again nonintegral. The other fork corner has identical contacts after white-path reversal. Thus none of their fork units is deleted when $I_E=1$. All eight other patterns were excluded. Every affected quadrilateral therefore retains a full basic unit even when several active fork units are removed. No triangle is altered, and on a longer face removing one unit from its unit angle preserves nonnegativity. This proves the active-state saving one on every D_5+2E_8 vertex.

4.1.2 Inactive extreme-E8 consumers, with every D5 contact

At the extreme pair $(0,6)$ of one old E_8, let $W_E=\{1,2,3,4,5,7\}=D_6$; the whole old D_5 and second E_8 are white spectators. Suppose a quadrilateral has no other basic unit. Its new walls are orthogonal. Deletion 0 leaves D_7, so the complete zero-payment endpoint types for c are N, $S=3@1$, $T=3@7$. Deletion 6 leaves E_7, giving for d the types N or $L=3@5$. Any S,T,L side is orthogonal to D_5: joining the old factors would give an unlisted rank-thirteen finite tree with two forks. An N side may contact D_5 by long-leaf 3 or short-leaf 3, with projection square $\delta=0,1,5/4$. If both do, only distinct long/short attachments are finite, with mutual projection $C=1/2$. This follows by finite D/E classification; two short attachments give an affine $(2,2,2)$ tree, and repeated leaves give two forks. Thus these statements include every spectator contact, not only a previously selected orthogonal configuration.

The needed old inverse entries are $J_{00}=4, J_{01}=7, J_{07}=5, J_{11}=14, J_{77}=8$, $J_{55}=6, J_{56}=3, J_{66}=2, J_{06}=2$. Eliminating W_E gives old-black matrix $\left(\begin{smallmatrix}1/2 & -1/2\\ -1/2 & 1\end{smallmatrix}\right)$. For N opposite squares X,Y, write $a=2-\delta_c, b=2-\delta_d$. Conjugate positivity gives $0\le X'\le a/4$, $0\le Y'\le b/2$. For the field-valued opposite products α of S,T and β of L, it gives respectively $\alpha'\in[3/2,2],[1,3/2]$, $\beta'\in[1,2]$; these are the roots of the height polynomials $12-14\alpha+4\alpha^2$, $6-10\alpha+4\alpha^2$, $4-6\beta+2\beta^2$.

The corank equations are

S,L	$(\alpha-1)\beta=1$
T,L	$(1-2\alpha)(-\beta)=2$
N,L	$(2X-a)(-\beta)=a$
S,N	$(1-\alpha)(Y-b)=b/2$
T,N	$(1-2\alpha)(Y-b)=b.$

Every displayed unit factor is physically positive by opposite indefiniteness. The unit proof above gives the following complete short intervals (negative conjugates are essential):

UV	U' interval	U
1	$[-1,-1/2]$	φ
1	$[-2,-1]$	φ^{-1}
2	$[-2,-1]$	$\varphi^{-1}, 2\varphi$
3	$[-3,-3/2]$ or $[-2,-1]$	$\varphi^{-1}, 3\varphi$.

They follow from $U = p^a\varphi^r$, $a = 0, 1$, r odd; the adjacent odd exponents already lie outside the interval. Applying them gives all mixed-endpoint obstructions:

	$\delta = 0$	$\delta = 1$	$\delta = 5/4$
N, L	$\beta = -\varphi^{-1}$	$X = \varphi^2/2 \notin \mathcal{O}$	$8X - 3 = \varphi^{-1}$ or 3φ
S, N	$\alpha = 1 - \varphi$	$2(1-\alpha)(Y-1) = 1$	$(1-\alpha)(8Y-6) = 3$
T, N	$1 - 2\alpha = \varphi^{-1}$ or 2φ	$1 - 2\alpha = \varphi^{-1} < 1$	$1 - 2\alpha = \varphi^{-1}$ or 3φ.

Entries give either a forbidden finite product in $(-1, 0)$, a positive opposite product, a nonintegral coordinate, or the displayed modulo-two contradiction. The S, L row gives $1 - \alpha = \varphi$, again forbidden; T, L gives the same positive/nonintegral alternatives as its table row. Thus every pattern except N, N is excluded.

For N, N the complete equation is

$$4XY - 4bX - 2aY + ab - 4C\sqrt{XY} - C^2 = 0, \qquad X > a/2, \quad Y > b. \tag{1}$$

With at most one D_5 contact, $C = 0$ and it factors as $(2X - a)(Y - b) = ab/2$. The nonzero contact pairs $(\delta_c, \delta_d) = (1, 0), (5/4, 0)$ give respectively nonintegral $X = \varphi^2/2$ or $8X - 3 = \varphi^{-1}, 3\varphi$. The pairs $(0, 1), (0, 5/4)$ give respectively $2(X - 1)(Y - 1) = 1$ or $(X - 1)(8Y - 6) = 3$, impossible modulo two. For both contacts $C = 1/2$ and the Cartan cycle gives $t = \sqrt{XY} \in \mathcal{O}$. The long/short orientation gives $8t^2 - 4t + 1 - 6X - 4Y = 0$, impossible modulo two. The short/long orientation gives $8t^2 - 4t + 1 - 8X - 3Y = 0$. Here $X' \le 3/16$ excludes every finite X from the two-cycle list; if $X, Y \ge 4$, the left side of (1) is $\ge 69/2 > 0$. Hence Y is finite, and $Y > 1, Y' \le 1/2$ leaves only $Y = \varphi^2$. Substitution gives discriminant $20 + 8s$ of norm 80, not a square. This excludes the last D_5 contact. For no contacts, $(X - 1)(Y - 2) = 1$ and the short unit intervals give $X = Y = \varphi^2$. Thus the two new sides are exactly the actual E_8 triangle walls F_0, F_6 of the preceding table: their other positive replacements close the designated triangle bundles.

Consequently, when $I_E = 0$, an extreme-E8 quadrilateral must have another basic unit. That unit cannot be an actively deleted D_5 fork unit: the old whole D_5 is white and cannot join the white D_6 into a finite component with two forks. Its component after adding the marks has rank at most seven, hence cannot turn into an E_8 at another $D_5 + 2E_8$ corner. There it remains the whole white D_5, so the black marks cannot be its fork leaves. This proves simultaneous safety of inactive E8 deletions and active D5 deletions. Inactivity also excludes triangles at that extreme pair; longer faces need only nonnegative weights. Removing its nonnegative bonus is safe. The two whole E8 copies have the same context and bonus b, so the inactive saving is $2b$, while the active saving is one. Imposing both physical-state rows gives exactly $W - \min(1, 2b)$.

4.1.3 The H4 deletion and all four source capacities

Number H_4 as $0-1-2-3$ with bonds $5, 3, 3$, and its whole A_1 as node 4. Remove t from the extra five-edge unit at the target $A_1 + H_4 + 2E_8$. A triangular face at $(0, 1)$ has third side c orthogonal

to both E8 factors. A contact label at least six to an old nonisolated root is incompatible with its retained old neighbor, except possibly at marked leaf 0. That exception gives a retained ordinary unit at the other triangle corner, and the old ordinary five-edge unit gives the second. A high bond to the whole A_1 forces every H4 contact zero, giving height $(2\cos(\pi/m))^2/2 - 2 < 0$. Thus every potentially deficient case lies in the exact domain $\{2,3,4,5\}^5$.

The bounded test has no unspecified geometric filter. Let g_i be the five contact magnitudes, $J = J_{H_4} \oplus (2)$ and $H = g^T J^{-1} g - 2$. For all 4^5 label vectors test both old-root replacements by finite classification, $H > 0$, field membership of each old-path cycle, $H' \leq 0$, and all connected affine principals containing c. Subsets containing a 5 need no affine test, since no affine Coxeter diagram has such a bond. There are 46 finite-corner eligible vectors, four with nonpositive height, eight failing cycle membership, and 34 remaining necessary vectors. Evaluating all three baseline corners after deleting every H4 five-edge extra at once leaves exactly the following deficient rows:

H4 contact	A1 contact	H	source corner omitting 0
3@3	0	$1+s$	$A_1 + A_4 + 2E_8$
3@3	3	$3/2+s$	$A_5 + 2E_8$
3@3	4	$2+s$	$B_5 + 2E_8$
4@3	0	$4+2s$	$A_1 + B_4 + 2E_8$.

The full label-vector certificate records every eligible vector, its exact inverse/height tests and its three payments; this small finite table is part of the computer-assisted data, not a high-label search. In these four rows the original payments are $(2,0,0)$, and only the original five-edge extra is removed. Restore t at the source angle marked by old 1 and c. Its baseline is zero, so the sum is again two. All other vectors remain paid even after every simultaneous deletion. A triangle with only some deletions is therefore safe.

The possible source pairs are intrinsic: the endpoint pair in A_4, the two pairs $(0,3),(1,4)$ in a source A_5 path, the endpoint pair in B_4, and $(1,4)$ in B_5 with four-bond $(0,1)$. An angle belongs to one two-face, so restoring it once gives capacity one in each type except the apparent A5 capacity two. That last pair cannot coexist. Its two restorations require actual walls x, y with single contacts $-\varphi$ at the opposite endpoints of the source A_5, both orthogonal to the two whole E8 factors. The full 21-root old positive space has one-dimensional timelike complement, so their heights have the same positive sign. The A5 inverse endpoint entries are $5/6, 1/6$, giving

$$H = 5\varphi^2/6 - 2 = (-9+5s)/12 > 0, \qquad x \cdot y = \varphi^2/6 - H = (3-s)/3 > 0.$$

This violates acuteness. Thus every one of the four source capacities is one, proving the $W + t$ rows of Proposition 4.2.

Every affected quadrilateral retains the ordinary unit at its modified H4 angle. Restorations are nonnegative; longer faces remain nonnegative because $t \leq 1$ is removed from a two-unit angle. The old D5 fork subtraction never affects triangles. Its inactive E8 subtraction acts inside a white whole E8 in the four H4 patterns and so cannot affect these marked triangle angles. Conversely the H4 deletion preserves all ordinary units used in the simultaneous D5/E8 argument. This proves all face inequalities and all claimed priority allowances together. In particular the persistent-payment pruning in the contact queue remains valid. An active fork-deletion quadrilateral gets its alternative ordinary unit by the complete eleven-pattern argument. An inactive extreme-E8 deletion on a quad with no ordinary payment would force F_0, F_6 and hence activity, a contradiction. An H4 deletion keeps its ordinary unit. Thus the old zero-baseline native rows remain literal valid obligations; no adjusted numerical bonus is needed to justify a geometric prune.

4.2 The new same-hub pools

Lemma 4.3. *At a simple finite vertex with $k \geq 2$ whole copies of a rank-r type H, there are $r^2\binom{k}{2}$ planar angles whose marked roots lie in distinct copies. Assign each the nonnegative addition*

$$\frac{u_H(q)}{r^2\binom{k}{2}}.$$

The total new vertex charge is exactly $u_H(q)$. If $k < 2$ the family contributes zero.

Proof. Choose an unordered pair of components and one of the r roots from each. These are distinct unordered marked facet pairs and exhaust the family. Summing cancels the denominator. The full tuple q is a vertex invariant and is constant throughout this sum. □

We use $H = A_1, I_2(5), H_3$. Their per-angle denominators are $\binom{a}{2}$, $2b(b-1)$ and $9h(h-1)/2$. The three families are disjoint, and their nonnegative additions preserve paid triangles and long faces. Proposition 3.5 supplies actual whole components and actual counts when these pools pay an otherwise unpaid native quad. Its once-only new vertex cost is

$$\rho(q) = \mathbf{1}_{a\geq 2}u_{A_1}(q) + \mathbf{1}_{b\geq 2}u_{I_2(5)}(q) + \mathbf{1}_{h\geq 2}u_{H_3}(q).$$

All six special forests of Proposition 4.2 have $\rho = 0$.

5 Complete forest maximization and exact certificates

The full rank 21 alphabet has68 irreducible types:

$$A_1, \ldots, A_{21}; \quad B_2, \ldots, B_{21}; \quad D_4, \ldots, D_{21};$$
$$E_6, E_7, E_8, F_4, H_3, H_4, I_2(5), I_2(6), I_2(10).$$

Fix exact hub counts q and remove the four hub types from the residual alphabet. If $w_c(T)$ is the old additive cost of type T, define

$$U_c(0) = 0, \qquad U_c(s) = \max_T\{U_c(s - \operatorname{rk} T) + w_c(T)\}$$

on reachable ranks; rank one is omitted. Every recurrence choice appends an allowed component, and every nonempty multiset has a final component to remove. Hence arbitrary repetitions are covered exactly. The ordinary budget adds fixed hub cost and $\rho(q)$ once to $U_c(21-a-2b-3h-4d)$, never once per residual component.

Here are the full loop bounds and the correctness argument. Enumerate all nonnegative hub tuples (a, b, h, d) with $a + 2b + 3h + 4d \leq 21$. The capped tuple fixes the old additive costs; the exact tuple fixes the new pools. Evaluate the residual recurrence in increasing rank from 0 through $21 - a - 2b - 3h - 4d$. An empty maximum is an explicit unreachable state. The loops terminate because there are finitely many tuples,68 component types and at most 22 residual ranks. Each recurrence transition lowers residual rank by a positive integer. Induction on that rank proves that every computed value is attained by a residual multiset and every such multiset is bounded by a considered last-component transition. Thus the recurrence includes arbitrary repetitions and every reachable forest.

This conclusion concerns the fixed context. A presence bonus or a cross-component pool cannot be silently treated as a repeated item weight. Fixing the hub counts first is what makes the residual

cost additive. The physical exceptions below are then evaluated as separate offers in the same maximum.

Two ordinary budget offers are replaced to exclude exactly the physical decreased forests. If a residual forest contains an off-target type, reserve one such component and fill the remaining rank by U_c. The pure H4/E8 rank 20 possibilities are $5H_4$, $3H_4 + E_8$ and $H_4 + 2E_8$; the first two remain unchanged and are imposed explicitly with A1. The only pure D5/E8 rank 21 solution is $D_5 + 2E_8$, imposed in both states. All four positive sources are also evaluated with $+t$ even if they were not ordinary maximizers. No forest is discarded or assigned an independently favorable physical state.

Exact certificate 5.1. *The rational vector in* `results/fresh_compact21_self_hub_exact40_reconstruction_20261018.json` *has these exact properties:*

1. *All bonuses are nonnegative and* $t = 14030552/21687803 \in [0, 1]$.
2. *All1979 native faces and 102917 common spectator cases have quadrilateral payment at least* $60007431727/60000000000 > 1$.
3. *All68 allowed finite types and 729 exact hub-count tuples, including the priority exceptions, have total vertex allowance at most 40. The maximum40 is attained by* D_{21}.

The exact engine retains53495 distinct enlarged native inequalities. It appends new coefficients before deduplicating; projecting to old columns recovers51443 old classes. It reconstructs the 510 old support rows, but full coverage uses the 729 tuples and complete recurrence, not just those selected rows. There are243 capped groups and 1659 auxiliary rank states. Numerical auxiliary values are discarded.

A numerical optimizer was used to find a candidate. The final vector is rational, and every face payment and forest allowance is checked exactly. Neither optimizer convergence nor a floating-point tolerance is a premise of the proof.

Independent checker `scripts/check_compact21_self_hub_primal_independent_20261018.py` reconstructs the old orbit coefficients, new selectors, all pool counts, and the complete recurrence without importing the new model or producer. The recorded independent run verified the stated minimum payment and exact maximum40 over850 final forest offers. Its ledger is `results/fresh_compact21_self_hub_exact40_independent_check_20261018.json`. Source hashes identify the precise checked data; they do not establish the geometric completeness propositions. To evaluate the attached vector directly, the array `positive_bonus_values` gives pairs (column, rational value), with every omitted column zero. Its 93045 old columns are decoded by the `variable_order` records in `results/fresh_sqrt5_dimension21_a4_source_orbits_20261016_cap2.json`: each record specifies the orbit just defined and its four capped-context coordinates. The arrays `pool_variable_order` and `positive_pool_values` similarly specify the exact-context same-hub coefficients; `transfer_value` is t. The attached finite data therefore determine every marked-angle value, not only the maximum or a list of selected support inequalities.

The face part of that checker has a similarly precise domain. For every saved final native row, it enumerates all the common spectator tuples specified after Proposition 3.5, determines the four actual marked orbits and full hub multiplicities, and evaluates the rational weights. The finite row and tuple bounds prove termination. Completeness for actual faces is supplied by Propositions 3.3–3.5, not by the reported number of successful row evaluations.

Proposition 5.2. *No compact arithmetic hyperbolic Coxeter polytope with actual ground field* $\mathbb{Q}(\sqrt{5})$ *has dimension 21.*

Proof. Apply the angle rule to each actual finite vertex forest. Proposition 3.5 and Certificate 5.1 pay every otherwise unpaid quadrilateral. The baseline triangle theorem and Proposition 4.2 pay every triangle. Every other two-face needs only nonnegativity. The exact forest certificate bounds each actual vertex by40. This contradicts Lemma 2.1. □

6 The nongolden field obstructions

We now prove the two remaining internal field arguments used in the dimension21 reduction. Write $F = \mathbb{Q}(\sqrt{2}, \sqrt{5})$ and $K_7 = \mathbb{Q}(\cos(2\pi/7))$. All roots in this section have square one.

Lemma 6.1 (The field of an indefinite principal block)**.** *The determinant of a Lannér principal Gram matrix generates the actual arithmetic ground field K.*

Proof. Expand the determinant into permutation cycles. Each term is a rational multiple of cyclic Gram products, so the determinant lies in K; individual normalized cosine entries need not lie in K. The determinant is negative physically and nonzero. Every nonidentity embedding of K extends to the totally real entry field, where arithmeticity makes the principal matrix positive semidefinite. Its determinant cannot become zero under an embedding, and is therefore positive. If its generated subfield were proper, separability and total reality would give a nonidentity embedding of K fixing that subfield. It would preserve the negative determinant, a contradiction. □

Here is the common geometric obstruction. Choose minimal indefinite pieces in the two opposite extensions of an actual quadrilateral, which are indefinite by Lemma 3.1: ellipticity would make opposite sides meet, and an affine principal block contradicts compactness. Each minimal piece contains both opposite black roots, because the white support with either single black root is positive definite. Suppose that every such piece has order two or three. If no adjacent black pair has distance at most two in its finite corner, both pieces must be triangles on distinct white roots u, v. Indeed every cross black–black contact gives a distance-one corner, and a cross black–white contact gives a distance-two corner through one of the triangle's black neighbors. A shared white root gives the same contradiction. If either piece had order two, these restrictions would make the two indefinite pieces orthogonal. Thus their only possible connection is the nonright white bond uv.

The four black roots form a positive-definite direct sum of two opposite pairs. Eliminate those pairs. If each white projection cost is greater than two, the residual diagonal entries are less than -1. The residual off-diagonal is the original uv product, whose square is less than one because the white diagram is positive definite. The residual matrix is negative definite, giving two negative directions in a Lorentz principal matrix. This is impossible. Additional white roots cannot repair this principal obstruction.

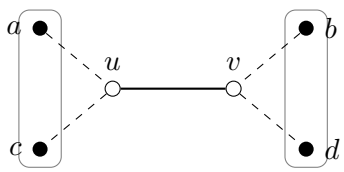


Figure 1: The common six-root configuration after the short-corner exclusions. The boxes are the two positive opposite-pair blocks to eliminate; their internal products are suppressed. Dashed lines show allowed white contacts, with a nonempty subset chosen on each side. The only cross connection is the nonright bridge uv; all omitted cross contacts are zero.

Proposition 6.2 (The distinguished-bond argument)**.** *A compact arithmetic21-polytope cannot have an actual ground field K which is not isomorphic to a subfield of F and does not contain K_7.*

Proof. A Lannér diagram of order at least four has no bond of label at least six: a neighboring nonright bond would give a proper three-node determinant at most $1 - 3/4 - 1/4 = 0$. Dotted or parallel bonds already fail a proper two-node test. All remaining entries belong to F, contrary to Lemma 6.1. The same field argument excludes triangles with all labels at most five. Label7 is impossible since its two-cycle generates K_7. Every remaining triangle consequently has either a bond of label at least nine, or a label6 or8 with both other bonds nonright. In the latter case a right triangle would again have determinant in F.

Call such a bond distinguished. In any finite diagram it is an entire rank-two component: a further nonright contact violates the three-node positive-definiteness test above. In the quadrilateral configuration just described, the white bridge therefore forces both distinguished bonds onto the opposite black pairs. If c is their positive contact magnitude and p, q are the nonnegative white contact magnitudes, the projection cost is

$$C = \frac{p^2 + q^2 + 2cpq}{1 - c^2}.$$

For label at least nine, one nonright contact gives $C > 2$: $\cos^2(\pi/9) > 7/8$. To verify this strict bound, put $z = 2\cos(2\pi/9) > 1$. The polynomial $z^3 - 3z + 1$ is increasing for $z > 1$ and is $-1/8$ at $3/2$, so $z > 3/2$. For a distinguished label6 or8, both contacts are nonright and $C \geq 1/[2(1-c)] > 2$, since $c \geq \sqrt{3}/2 > 3/4$. The common obstruction proves a short marked corner on every quadrilateral.

Give one unit to distance-at-most-two marked pairs and one additional unit to an entire $I_2(m)$ pair with $m \geq 6$. A triangular face has a minimal indefinite block containing all three black nodes: deleting one leaves a finite corner. The full star has a negative direction, as projection off its white support gives the normals bounding its actual compact hyperbolic triangle. The minimal block is therefore exactly that black triangle. Its two nonright bonds give payment at least two. Quadrilaterals pay at least one as proved above. For a connected finite tree the distance-two pair count is

$$(r - 1) + \sum_v \binom{\deg v}{2}.$$

It is $2r - 3$ on a nontrivial path and $2r - 2$ on a D/E tree; A_1 costs zero. The additional dihedral unit gives cost two, also $2r - 2$. Every rank21 forest thus costs at most40. Lemma 2.1 gives the contradiction. □

6.1 The exact biquadratic field

Assume now that the actual ground field is F, not a proper subfield. The published arithmetic Lannér classification in [2, Section3.1, Figure1 and formula(4)] puts the ground fields of orders four and five in $\{\mathbb{Q}, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{5})\}$. Lemma 6.1 excludes those orders here. Order two remains the full, unsampled family

$$\tau \in \mathcal{O}_F, \quad \mathbb{Q}(\tau) = F, \quad \tau > 4, \quad 0 < \sigma(\tau) < 4 \quad (\sigma \neq \mathrm{id}),$$

with unit-normal product $-\sqrt{\tau}/2$. The preceding quadrilateral argument eliminates an order-two minimal piece before using a triangle list, so no bounded dotted-parameter search is needed in this field.

Put $t_m = 2\cos(\pi/m)$ and $d_m = t_m^2$. The condition $d_m \in F$ makes $\varphi(m)$ divide eight for $m \geq 3$. Every prime divisor of m is2,3 or5, with exponents at most4,1,1. Testing those finitely many products gives

$$2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30.$$

Labels12 and24 contain $\mathbb{Q}(\sqrt{3})$, not a subfield of F. The real cyclotomic fields for15,16,20,30 are cyclic quartic: the respective classes2,3,3,7 have order four in the unit group modulo sign. They cannot equal the biquadratic field F. The complete finite alphabet is therefore

$$\mathcal{L} = \{2, 3, 4, 5, 6, 8, 10\}, \qquad (d_m)_{m\in\mathcal{L}} = (0, 1, 2, \phi^2, 3, 2+\sqrt{2}, 2+\phi), \quad \phi = (1+\sqrt{5})/2.$$

Exact certificate 6.3 (The finite triangle calculation)**.** *Among the* $\binom{9}{3} = 84$ *unordered triples with repetition from* $\mathcal{L}$*, the complete arithmetic Lannér triangles with actual field* F *are* $(2, 5, 8)$ *and* $(4, 5, 5)$*. The successive tests give:*

First failed test or surviving class	*Number*
Not a compact hyperbolic triangle	*13*
Length-three cyclic product outside F	*37*
Proper ground subfield	*28*
Nonpositive conjugate determinant	*4*
All tests passed	*2*

Here is a complete exact algorithm for this finite proposition. Enumerate $l \leq m \leq n$ from $\mathcal{L}$, reject reciprocal sum at least one, and seek the nonnegative element $P = \sqrt{d_l d_m d_n}$ in F. Arithmetic is rational in the basis $1, \sqrt{2}, \sqrt{5}, \sqrt{10}$; no floating sign decision occurs. For completeness, square-root membership in a quadratic extension $L(\sqrt{d})$ reduces to square-root membership in L. To square $y + z\sqrt{d}$ to $a + b\sqrt{d}$, solve

$$y^2 + dz^2 = a, \qquad 2yz = b.$$

If $b = 0$, test both a and a/d for squares in L. If $b \neq 0$, enumerate both possible roots $r^2 = a^2 - db^2$ in L, test $y^2 = (a + r)/2$ with $y \neq 0$, and put $z = b/(2y)$, verifying the square. Conversely every root has exactly this form, so the test is complete. Apply it first to $F = \mathbb{Q}(\sqrt{2})(\sqrt{5})$ and then to $\mathbb{Q}(\sqrt{2})$; at the rational base, numerator and denominator must be nonnegative integer squares. The tower has fixed depth and every branch is finite.

For each successful P, the square-two determinant is

$$\Delta = 2(4 - d_l - d_m - d_n - P).$$

Test the four sign changes of $\sqrt{2}, \sqrt{5}$ on the generators d_l, d_m, d_n, P. Their common stabilizer is trivial precisely when they generate F, by Galois correspondence. Finally require $\sigma(\Delta) > 0$ at the other three embeddings. Exact signs in each quadratic tower follow by separating equal-sign summands or squaring opposite-sign summands; for example $\operatorname{sgn}(A+B\sqrt{d}) = \operatorname{sgn}(A)\operatorname{sgn}(A^2 - dB^2)$ when A, B have opposite signs. All cyclic products tested are integral because they are products of algebraic integers $2\cos(\pi/m)$. Proper two-node minors are positive at every embedding. These tests are consequently the arithmetic Gram criterion for these triangles. The fixed loops terminate, cover every triple, and return exactly the displayed partition and two survivors. The standalone rational implementation and its saved exact records are `scripts/fresh_sqrt2sqrt5_lanner_catalogue_20261016.py` and `results/fresh_sqrt2sqrt5_lanner_catalogue_20261016.json`.

Proposition 6.4. *There is no compact arithmetic21-polytope with actual ground field* F*.*

Proof. In the two-white quadrilateral configuration, a triangle whose opposite black bond has label m and white contacts p, q has projection cost

$$C(m; p, q) = \frac{d_p + d_q + P}{4 - d_m}.$$

The five possible positions from Certificate 6.3 give

Triangle	Opposite bond	Cost
$(2, 5, 8)$	2	$(7 + 2\sqrt{2} + \sqrt{5})/8$
$(2, 5, 8)$	5	$1 + \sqrt{2}/2 + \sqrt{5}/5 + \sqrt{10}/10$
$(2, 5, 8)$	8	$3/2 + 3\sqrt{2}/4 + \sqrt{5}/2 + \sqrt{10}/4$
$(4, 5, 5)$	4	$3/2 + 3\sqrt{2}/4 + \sqrt{5}/2 + \sqrt{10}/4$
$(4, 5, 5)$	5	$2 + \sqrt{2} + 3\sqrt{5}/5 + 2\sqrt{10}/5$

The last four costs are greater than two. For the second row use $\sqrt{2} > 7/5$, $\sqrt{5} > 2$ and $\sqrt{10} > 3$, giving $1 + 7/10 + 2/5 + 3/10 = 12/5 > 2$ strictly; the remaining rows already exceed two from their first positive terms using $\sqrt{2} > 1, \sqrt{5} > 2$. The first position cannot occur: its white root meets a label8 bond and the required nonright white bridge in a finite corner, whose three-node determinant is at most $1 - \cos^2(\pi/8) - 1/4 < 0$. The common Schur obstruction now proves a short corner on every quadrilateral, with arbitrary other supporting whites retained. Use only the distance-two base weights. Each of the two possible black triangles has at least two nonright bonds, and so pays at least two. The same tree count bounds every rank21 vertex by40, again contradicting Lemma 2.1. □

7 The thirteen fields and final composition

Let $K_m = \mathbb{Q}(\cos(2\pi/m))$. The dimension-at-least14 field restriction [1, 2] and the inherited complete field proofs give the following dimension 21 branches. The shorter list valid only from 22 is not used at21.

Actual field	Dimension21 exclusion
$\mathbb{Q}$	Simplest-type arithmeticity and Meyer isotropy
$\mathbb{Q}(\sqrt{2})$	Complete all-white bound20 proof
K_7	Complete cubic-seven bound20 proof
$\mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{6}), \mathbb{Q}(\sqrt{2}, \sqrt{3})$	Proposition 6.2
$K_9, K_{11}, K_{15}, K_{16}, K_{20}$	Same proposition, field checks below
$\mathbb{Q}(\sqrt{2}, \sqrt{5})$	Proposition 6.4
$\mathbb{Q}(\sqrt{5})$	Proposition 5.2

Here are the precise internal theorem interfaces behind Proposition 1.2. The all-white $\mathbb{Q}(\sqrt{2})$ theorem excludes dimension21 using its complete arbitrary-white quadrilateral closure and strict weight comparison; both parts are required, not just the saved weight vector. Their common TeX exposition, including the all-size attachments, is Section4 of the preserved technical supplement `manuscripts/compact_arithmetic21.tex`. The supporting exact proof reports are `results/fresh_sqrt2_quad_exhaustion_20261015_review.md` and `results/fresh_sqrt2_weight_bound_20261015_review.md`. The cubic-seven bound20 theorem is the complete simultaneous-star rank proof in `results/fresh_compact_global_bound_20261015.md`.

For the eight branches of Proposition 6.2, the only subfields of F are $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{10})$ and F. The three displayed square-root fields are not among them. K_9, K_{11} have degrees three and five; K_{15}, K_{16}, K_{20} are cyclic quartic rather than biquadratic. None embeds in F. All eight have degree indivisible by three, except K_9. The cyclotomic fields of coprime conductors7 and9 have product degree $\varphi(63) = \varphi(7)\varphi(9)$, so their intersection is rational and K_9 does not contain K_7 either. This checks both field hypotheses for every branch. Proposition 6.4 uses exact field equality and is not applied to the golden proper subfield.

The original supporting derivations and exact records remain preserved as `results/fresh_arithmetic_high_label_20261016.md` and `results/fresh_sqrt2sqrt5_lanner_catalogue_20261016.md`; their geometric arguments have now been given here. The rational branch uses simplest-type arithmeticity and Meyer: the rational indefinite form in this rank is isotropic, so its arithmetic quotient cannot be compact.

For dimensions22 through29, the published higher-dimensional field restriction leaves $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$, K_7. The first and third have the bound20 just stated; the golden bound21 uses the complete native face domains and shared H3-hub rule in `results/fresh_sqrt5_h3_hub_coverage_20261016.md`. The strict dimension-dependent h-vector threshold and the full forest recurrence exclude each of these finitely many field/dimension branches. Together with the published compact29 theorem this proves the first assertion of Proposition 1.2. The thirteen dimension21 branches above prove its second assertion, without applying the dimension22 field restriction at dimension21. The preserved composition proofs are `results/fresh_compact21_single_field_20261016.md` and the technical supplement cited in the introduction. Their binders check dependency and branch coverage; they do not replace these component theorems or reprove them by hashing their certificates.

Proof of Theorem 1.1. Proposition 1.2 excludes every dimension at least22. At dimension21 its thirteen-branch reduction leaves only the golden field. Proposition 5.2 excludes that field as well. Hence every compact arithmetic chamber has dimension at most 20. □

8 Reproducibility and verification boundaries

The proof's finite objects are exact field/signature domains, literal contact maps, finite-diagram orbit coefficients, native face payments and forest maxima. Their explicit source and data pins are carried by the rational reconstruction certificate. Reproduction needs no optimizer:

```
python3 -B \
  scripts/fresh_compact21_self_hub_exact40_reconstruction_20261018.py \
  --verify
python3 -B \
  scripts/check_compact21_self_hub_primal_independent_20261018.py \
  --candidate \
  results/fresh_compact21_self_hub_exact40_reconstruction_20261018.json
```

Run these as separate sequential batches through `scripts/run_bounded_research_check_20261018.py`, whose live watchdog adds the 59-second cap,1GiB process-tree RSS limit and disk/RAM gates. Each checker owns the shared exclusive lock. The first reconstructs the saved candidate and checks every enlarged face and forest; the second uses a separate implementation. Large inherited queue replays are documented in the bound21 proof map and must also be scheduled in bounded sequential batches. New final composition/hash checks bind their existing certificates; they do not freshly regenerate those classifications.

The remaining validation boundaries must be explicit. Published structural theorems may be assumptions in a formal interface. Internal whole-star coverage, incidence priority arguments and exact certificate logic must be formalized rather than silently imported as published axioms. A Lean theorem assuming all field bounds would certify only the last composition. The current result is an internally reviewed computer-assisted proof with rational scripts and an independent saved-primal evaluation; complete geometric Lean verification and external review are separate tasks. No new high-dimensional chamber or sharpness assertion follows.

A Exact finite data and proof supplements

The finite classifications in this paper use attached exact records, with the mathematical domains and rejection tests stated above. Their relevant proof and checker families are as follows. The listed stems identify JSON data under `results/` and solver-free checkers under `scripts/`. Written reports or separately named reviews provide the accompanying proofs and replay records. In particular, the triangle-coupling proof is in `results/review_compact21_d5_two_e8_triangle_coupling_20261018.md`. Full immutable dependency records remain in the reproduction guide rather than being printed as hash lists here.

- The complete Lannér and 76-marking inputs are `fresh_sqrt5_lanner_catalogue_20261015` and `fresh_sqrt5_triangle_weights_20261015`. A marking record has the literal graph labels, three black nodes, complementary white set, three finite corner graphs and their exact payments. The first catalogue has its separate exact arithmetic and topology replay.
- The initial union and 24 Schur families are `fresh_sqrt5_quad_exhaustion_20261015` and `fresh_sqrt5_dotted_parameters_20261015`. Their schema and bounded field domains are those in Proposition 3.3. The finite and dotted queues, their singular stopping obligations and literal marked maps are independently evaluated in the two queue review reports cited after Proposition 3.5.
- The active D5 proof uses `fresh_compact21_d5_fork_quad_triangle_priority_20261018`, its Lambert and one-basic-unit reports, and `fresh_compact21_d5_mixed_fork_quad_exclusion_20261018`. The checker domains are the 25 ordered pairs of the five finite endpoint types and the exact Schur polynomials and field obstructions displayed above, not a bounded distance search.
- The full inactive consumer is `fresh_compact21_extreme_e8_quad_consumer_completeness_20261018`. Its records retain the entire finite D5 two-root attachment domain, both orientations, short unit intervals and exact coefficient identities. The preliminary triangle types come from `fresh_compact21_d5_two_e8_triangle_coupling_20261018` and its separate written review; each root record has its full contact vector, height, two positive corner tests and conjugate field test.
- The 1024-vector H4 domain and four deficient rows are `fresh_compact21_h4_a1_triangle_priority_20261018`. Each eligible row records the five native contacts, exact height, cycle tests and all three modified triangle payments. The strengthened source capacity is `fresh_compact21_a5_h4_source_capacity_20261018`; its entire calculation is the two endpoint inverse entries and positive forced product given in the text.

The accompanying reports identify the finite domains and their exact execution records. The inherited field theorems remain the explicit companion interfaces of Proposition 1.2, with their technical proof homes in Section 7.

References

[1] E. B. Vinberg, *Absence of crystallographic groups of reflections in Lobachevskii spaces of large dimension*, Transactions of the Moscow Mathematical Society47 (1984),75–112. Original text: `https://www.mathnet.ru/php/getFT.phtml?jrnid=mmo&option_lang=rus&paperid=444&what=fullt`.

[2] V. V. Nikulin, *On ground fields of arithmetic hyperbolic reflection groups*, Introduction and Sections2,3, `https://arxiv.org/abs/0708.3991`.

[3] M. Belolipetsky, *Arithmetic hyperbolic reflection groups*, Lemma2.1 and Sections1,7, `https://arxiv.org/abs/1506.03111`.

[4] Exact internal sources, geometric reviews and immutable certificate interfaces, listed in `results/compact_arithmetic20_universal_composition_20261018.md`, `results/fresh_compact21_self_hub_exact40_reconstruction_20261018.md`, and `docs/compact_arithmetic21_proof_map.md`.